IMAGING COURSE · LESSON 3

Choosing the test

The best scan is the one that answers the clinical question with the least harm, delay and cost. Requests such as "CT head and neck please" waste radiation and time; requests such as "unilateral conductive loss with an attic retraction: is there cholesteatoma, and are the ossicles, tegmen and lateral canal intact?" get the right protocol and a useful report. This lesson matches the common ENT questions to the test that answers them.

AT A GLANCE

- Start from the clinical question, not the modality.
- Ultrasound first for thyroid, salivary glands, superficial lumps and nodes in the neck, with FNA.
- CT for bone (temporal bone, sinuses, fractures) and for acute infection; MRI for soft tissue, nerves and skull base spread.
- PET-CT for unknown primary, advanced stage and response after chemoradiotherapy (at about 12 weeks).
- Children: ultrasound and MRI before CT where possible.

1 CHAPTER ONE Start from the question

What do you need to know?

1.1 Bone, soft tissue, vessel or function?

Most ENT imaging questions fall into four kinds. **Bone and air** questions (sinus anatomy before surgery, ossicles, otic capsule, fractures, erosion by cholesteatoma) belong to **CT**. **Soft tissue** questions (the extent of a tumour, perineural or intracranial spread, the IAC, the labyrinth's fluid, cholesteatoma recurrence) belong to **MRI**. **Superficial lumps and glands** (thyroid, salivary glands, neck nodes, a lump in a child) belong to **ultrasound**, with fine-needle aspiration in the same sitting. **Function** questions (the swallow, aspiration, a pouch, metabolic activity of a tumour) belong to **contrast swallow, videofluoroscopy, nuclear medicine or PET-CT**.

COMMON ENT QUESTIONS AND THE TEST THAT ANSWERS THEM

QUESTION	FIRST TEST	WHY
Chronic rhinosinusitis, planning sinus surgery	Non-contrast CT sinuses (or cone-beam CT)	Bony anatomy, variants, skull base and orbit
Unilateral sinus opacification or suspected tumour	CT and contrast MRI	CT for bone; MRI separates tumour from secretions and shows orbit and dura
Orbital or intracranial complication of sinusitis	Contrast-enhanced CT; MRI if intracranial	Abscess, cavernous sinus, empyema
Chronic otitis media, cholesteatoma, conductive loss	Non-contrast thin-section CT temporal bones	Ossicles, scutum, tegmen, lateral canal, facial canal
Residual or recurrent cholesteatoma	Non-EPI diffusion MRI	Detects keratin down to about 3 mm
Asymmetric sensorineural loss or tinnitus	MRI of the internal auditory meatus (high-resolution T2 ± gadolinium)	Vestibular schwannoma and other CPA lesions
Pulsatile tinnitus	CT temporal bone with CT angiography and venography, or MRI with MRA and MRV	Paraganglioma, dehiscent jugular bulb, sinus stenosis, dural fistula
Facial palsy, atypical or not recovering	Contrast MRI from brainstem to parotid; CT for the canal	Tumour, perineural spread, fracture
Neck lump in an adult	Ultrasound with FNA, then contrast CT or MRI if malignant	Characterise, sample, stage
Neck lump in a child	Ultrasound	No radiation; most are nodes or congenital cysts
Thyroid nodule	Ultrasound (U or TI-RADS grading) with FNA	Risk grading and sampling
Salivary lump	Ultrasound with FNA or core; MRI for deep lobe or malignancy	Site, extent, facial nerve relation
Head and neck cancer staging	Contrast CT or MRI of primary and neck, chest CT; PET-CT for advanced disease	Local extent, nodes, distant spread

QUESTION	FIRST TEST	WHY
Unknown primary	PET-CT before panendoscopy	Finds the primary in a proportion
Response after chemoradiotherapy	PET-CT at about 12 weeks	Avoids unnecessary neck dissection (PET-NECK)
Dysphagia or suspected pouch	Barium swallow; videofluoroscopy for aspiration	Structure and function
Deep neck infection	Contrast-enhanced CT	Abscess versus phlegmon, airway, mediastinum
Facial or laryngeal fracture	Non-contrast thin CT	Fracture lines and displacement

CHAPTER TWO

2 The tests one by one

Strengths and limits.

2.1 Ultrasound

Ultrasound is quick, cheap, free of radiation and dynamic, and in skilled hands is the most accurate way to characterise thyroid nodules, salivary lumps and neck nodes and to guide FNA or core biopsy. A normal lymph node is oval with a bright fatty hilum; an abnormal one is round, loses its hilum, and may show necrosis, microcalcification or peripheral vascularity. Ultrasound cannot see behind bone or air, so it does not show the deep lobe of the parotid well, the retropharyngeal space, the larynx or the skull base, and it depends on the operator.

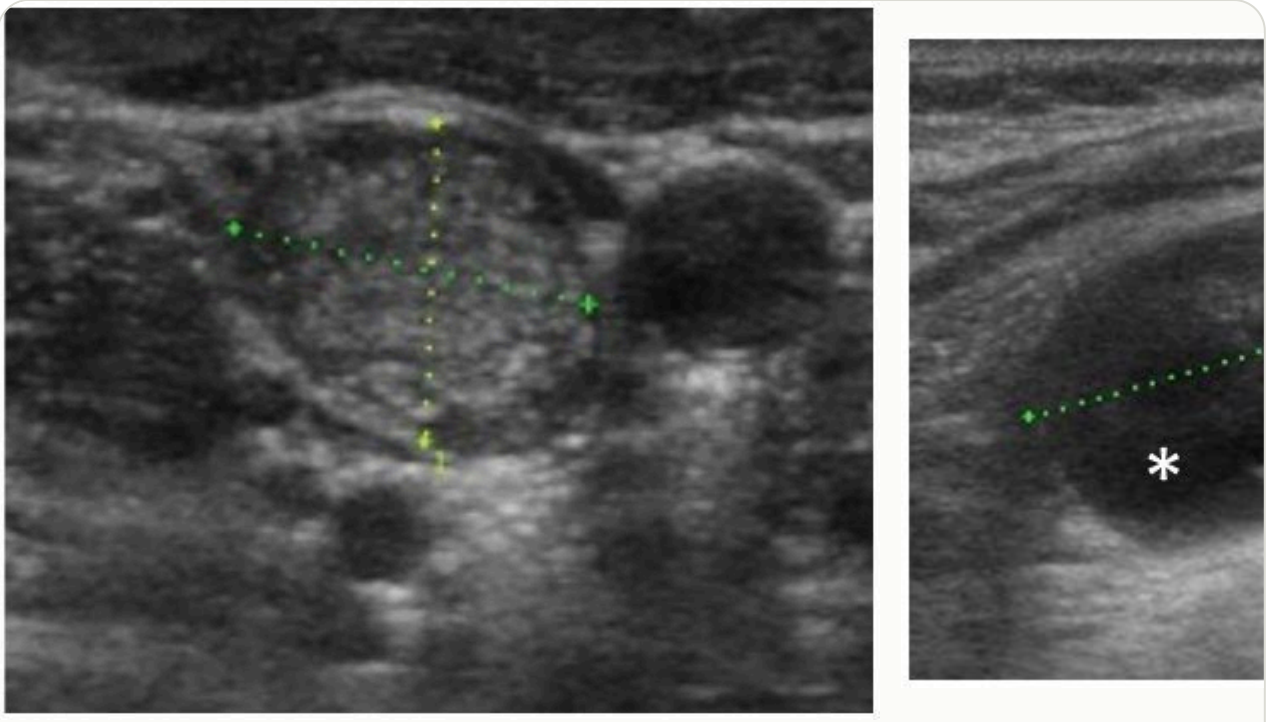


FIGURE 1 Abnormal neck nodes on ultrasound. Left: round, without a fatty hilum and with microcalcifications, typical of papillary thyroid cancer metastasis. Right: cystic change and loss of the hilum; fine-needle aspiration confirmed metastasis.

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2.2 Plain films

Plain radiographs have a small, specific role: a lateral soft-tissue neck film for a radio-opaque foreign body or for the thumb sign of supraglottitis (only if the airway is safe and it will not delay care), an orthopantomogram for dental disease, osteoradionecrosis and mandibular fractures, and a chest film. Sinus X-rays are no longer used to diagnose sinusitis.

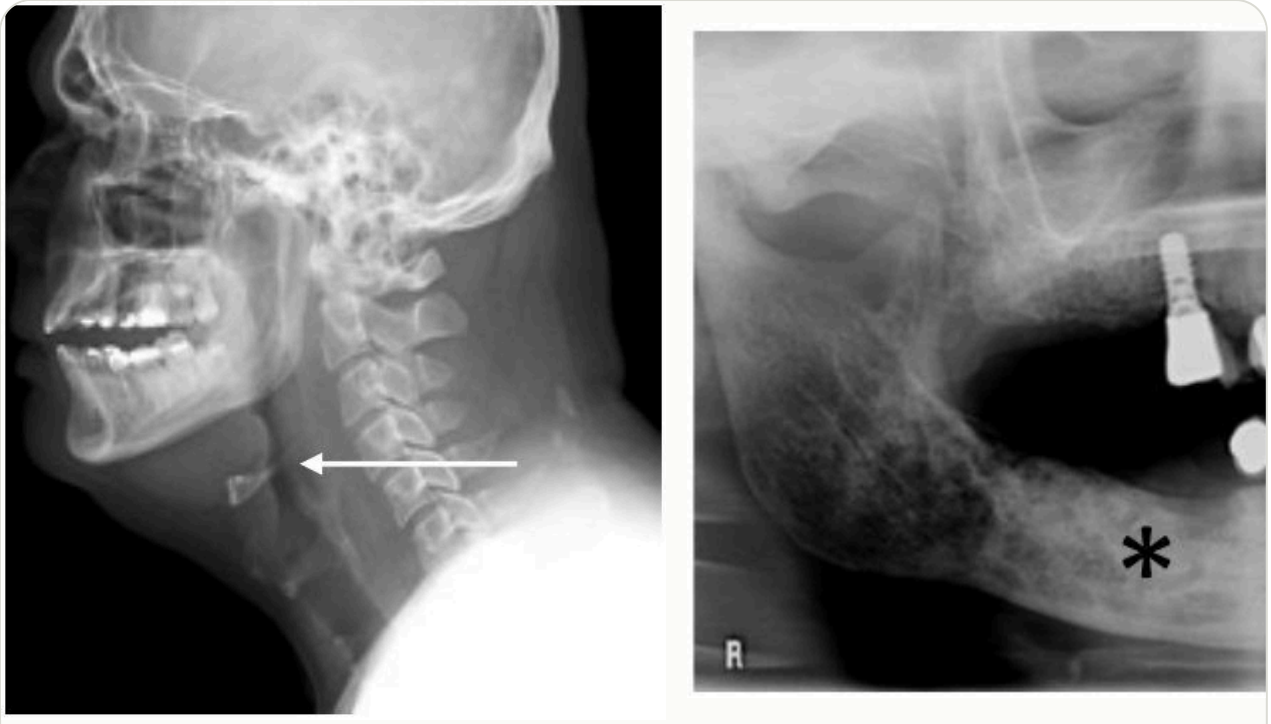


FIGURE 2 Plain films that still help. Left: lateral neck in acute supraglottitis, the swollen epiglottis narrowing the airway. Right: orthopantomogram showing osteoradionecrosis of the mandible.
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2.3 CT and MRI

CT and MRI are covered in their own lessons. In short, CT is fast, widely available and best for bone and air, acute infection and trauma; MRI is best for soft-tissue extent, nerves, the labyrinth, the skull base and recurrence, but is slow, sensitive to motion and contraindicated with some implants. In cancer of the skull base, sinuses and nasopharynx the two are complementary.

2.4 PET-CT

PET-CT combines a CT with fluorodeoxyglucose (FDG) uptake, which is high in most cancers but also in inflammation, infection, recent surgery, brown fat and normal lymphoid tissue (tonsils, adenoids). It is used for advanced-stage cancer and distant metastases, for the unknown primary, for suspected recurrence, and to assess response at about **12 weeks** after chemoradiotherapy: the PET-NECK trial showed that surveillance by PET-CT, operating only on residual uptake, was as safe as planned neck dissection and spared most patients the operation. Too early a scan gives false positives from radiation inflammation.

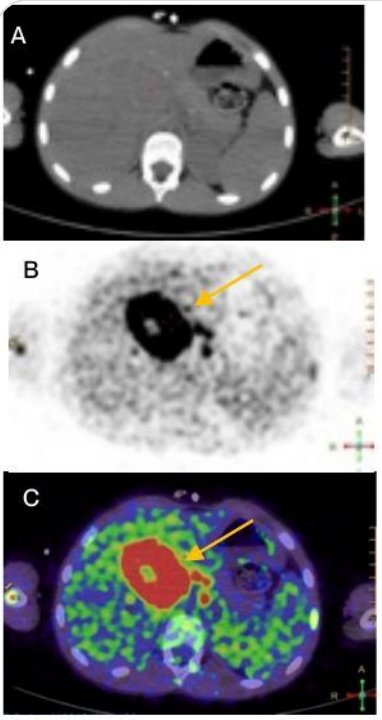


FIGURE 3 PET shows widespread disease in a child with lymphoma: the metabolic picture that CT alone would miss.

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2.5 Swallow studies

A **barium swallow** shows the anatomy of the pharynx and oesophagus: pouch, web, stricture, ring, achalasia and tumour. **Videofluoroscopy** (modified barium swallow) with a speech therapist shows the dynamics of the oral and pharyngeal phases and aspiration with different textures. Water-soluble contrast is used when perforation or a leak is suspected.

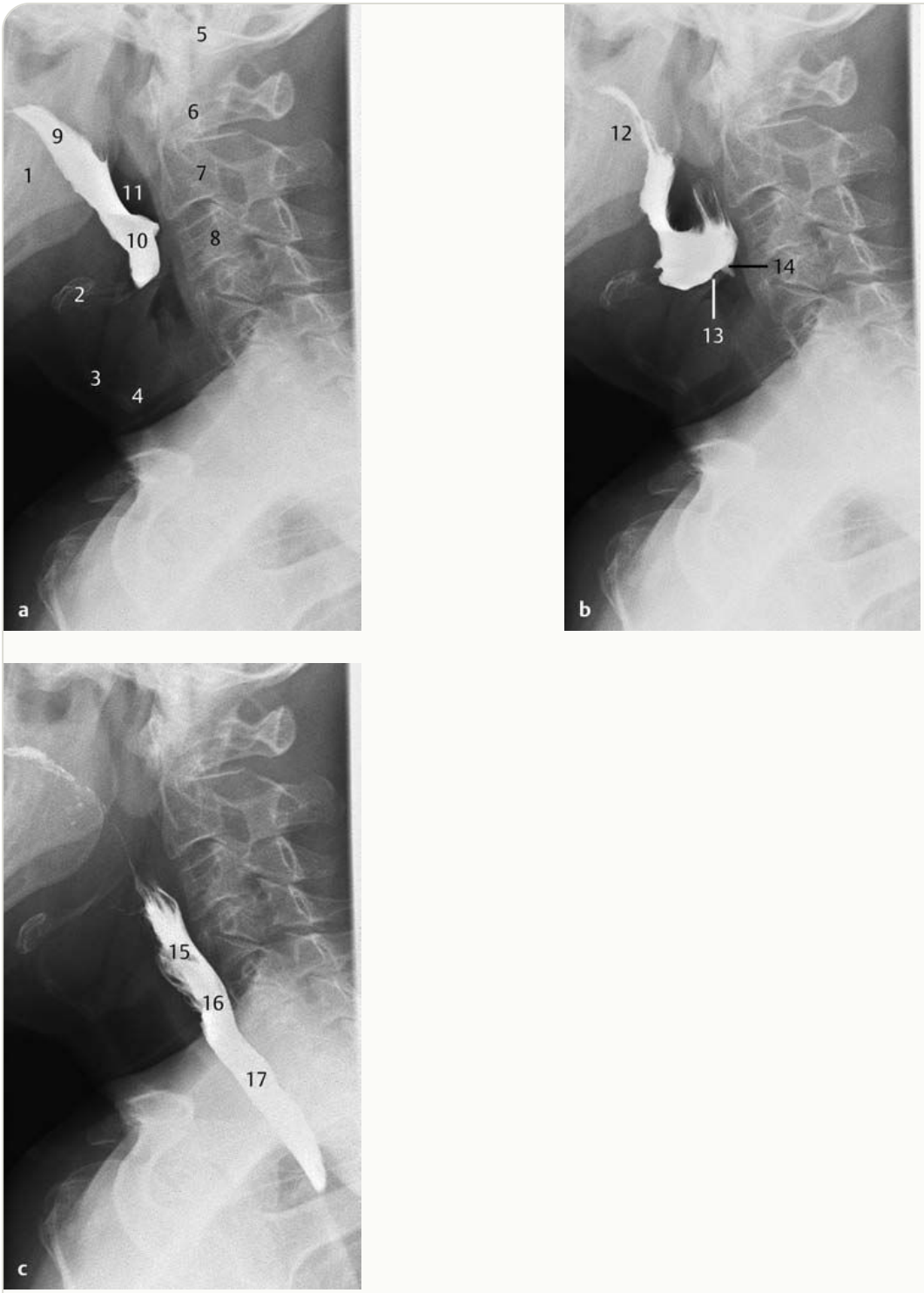


FIGURE 4 Contrast swallow images of the pharynx and oesophagus: a dynamic study shows both structure and transit.

Dunnebier, Imaging for Otolaryngologists

2.6 Angiography and nuclear medicine

Catheter angiography is now used mainly for treatment: embolisation of juvenile angiofibroma, paraganglioma and epistaxis, and balloon occlusion testing before

carotid sacrifice. **Nuclear medicine** offers sestamibi and SPECT-CT for parathyroid adenoma, iodine and technetium scans for the thyroid, and salivary scintigraphy for function.

3 CHAPTER THREE Safety and stewardship

Fewer, better scans.

3.1 Radiation, contrast and children

Each CT adds radiation, most important in children and for the lens of the eye. Ultrasound and MRI are preferred in children where they can answer the question; MRI in young children may need sedation or a "feed and wrap" approach in infants. Iodinated contrast is checked against allergy, renal function and planned radioactive iodine; gadolinium against renal function and pregnancy.

3.2 Writing a good request

State the clinical findings, the specific question, and anything that changes the protocol: suspected cholesteatoma (thin bone CT), pulsatile tinnitus (add angiography), cancer staging (contrast, include chest), cochlear implant or other metal (MRI safety). A precise question gets a precise answer.

CHOOSING THE TEST

1 What do you need to see?

- → **bone, air, ossicles, fracture**
  CT
- → **soft tissue, nerve, skull base, recurrence**
  MRI
- → **superficial lump or gland**
  ultrasound ± FNA
- → **swallow or aspiration**
  barium or videofluoroscopy

2 Is it urgent?

- → **airway, abscess, complication, trauma**
  contrast CT now
- → otherwise plan the best test

3 Is it a child?

- → prefer ultrasound or MRI
- → CT for bone questions and emergencies

4 Is it cancer?

- → primary and neck by CT or MRI
- → chest CT
- → PET-CT for advanced, unknown primary or response at 12 weeks

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Match the test to the question:** bone, soft tissue, superficial gland or function.
- **Ultrasound** features of an abnormal node.
- **PET-CT:** indications, false positives, and timing after chemoradiotherapy (PET-NECK).
- **Barium swallow versus videofluoroscopy.**
- **Imaging children** with the least radiation.
- **A good imaging request.**

4.2 Common pitfalls

COMMON PITFALLS

- Ordering a CT for a thyroid nodule or a child's neck lump instead of ultrasound.
- Requesting PET-CT too soon after radiotherapy.
- Ordering MRI for a fracture or ossicular detail.

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